GOVERNMENT OF INDIA
MINISTRY OF RAILWAYS
(RAILWAY BOARD)

No. E(GP)2026/2/5 (3522005) New Delhi, dated-As signed

The General Managers,
All Zonal Railways / Production Units / Training Institutes.

Sub.: Introduction of Centralized CBT/TBT for promotion to Group 'B' posts in Miscellaneous and Isolated Cadres – Notification of syllabus – reg.

Ref.: Board's letters of even number dated 23.01.2026 and 16.04.2026.

As already informed vide Board's letter under reference, Centralized Computer Based Test (CBT)/Tablet Based Test (TBT) is being introduced for promotion to Group 'B' posts in Miscellaneous and Isolated Cadres for the common vacancy cycle from 01.01.2027 to 31.12.2028. Accordingly, the syllabi for the cadres has been finalized.

2. The **cadre-wise syllabi for the Professional Subject**, along with the **common Establishment & Financial Rules syllabus** applicable to all Miscellaneous and Isolated Cadres for which Centralized Computer Based Test (CBT)/Tablet Based Test (TBT) is being introduced, are enclosed as Annexures for further necessary action.

3. It may also be noted that the Government has consolidated 29 labour laws into four comprehensive Labour Codes, namely, the Code on Wages, 2019, the Industrial Relations Code, 2020, the Code on Social Security, 2020 and the Occupational Safety, Health and Working Conditions Code, 2020. Accordingly, wherever the syllabi refer to the erstwhile labour Acts/Laws, the relevant provisions of the four Labour Codes, as applicable, may be read in place of such erstwhile Acts/Laws.

4. This issues with the approval of the competent authority.

Encl.: As above.

Digitally signed by Bubble Yadav
Date: 29-09-2026 14:28:29

(Bubble Yadav)
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Annexure-I

Syllabus for Establishment Rules

1. Organization of the Personnel Department in Railways objectives functions and policies of Personnel Department.

2. Recruitment and Training, Classification of Services, Recruitment in Different services, Railway Recruitment Boards & Railway Recruitment Cells, Compassionate Ground Appointments, Initial & In Service Training, Refresher Courses, Central Training Institutes, Training Centers in Zones, Divisions & Workshops, Training Modules for different posts, Training under Apprentices Act, Online Training, APARs.

3. General conditions of service in Railways, Seniority, Lien, Inter Railway & Inter Division transfers, Deputation, Promotion Policy & methods. Selection, Suitability, Trade Tests, Leave Rules, Pass Rules, Joining Time Reservation policy, HOER, Overtime, Payment of wages, current CPC Pay Rules, Advances in Railways.

4. Manpower planning, Rightsizing & Benchmarking, creation, extension and surrender of posts, creation of posts against new assets, different types of posts including workcharged posts.

5. The Railway Servants (Discipline & Appeal) Rules, 1968 and related instructions.

6. The Railway Services (Conduct) Rules, 1966 and related instructions.

7. Retirement benefits, qualifying service, pension, family pension, commutation gratuity, new pension scheme.

8. Staff welfare, SBF, Railway institutes, Railway schools, Ex-gratia payment, Incentive Bouns Scheme, Staff Grievances Redressal Mechanisms.

9. Industrial relations in Railways, recognized trade unions, industrial disputes. The Industrial Relations Code, 2020, PNM, PREM, JCM, Various Associations & Informal Meetings.

10. The Occupational, Safety, Health & Working Conditions Code, 2020. The Code on Social Security, 2020, Functions of Labour Enforcement Officers, Right to Information Act.

11. The scope of Information Technology in Railway e - office. HRMS, IPAS, LIMBS, ARPAN, CPGRAMS, ANUBHAV etc.

Syllabus for Financial Rules:

1. Parliamentary Control over Railway Finance, Public Accountability, Cannons of Financial Propriety.

2. Railway Budget - Budgetary terms, Types of Budgets, Budget cycle, Demand of Grants, Budgetary and Financial Reviews.

3. Rules of Allocation - Classification of expenditure - Control of expenditure - Responsibility Accounting - Performance Budgeting - Exchequer Control - Financial Results of Working lines.

4. Works Programme - Financial justification of Works - Surveys - Preparation of Estimates - Capital Budget - Control over Capital Expenditure - Reappropriation of Funds.

5. Financial control over Stores Expenditure - Purchase and Stores Keeping Procedure - Inventory Control and ABC Analysis.

6. Financial & Cost Control in Railway Workshops/Sheds/Units.

7. Rules and procedure relating to Tenders and contracts for execution of works and Procurement of Stores, M&P Programme and RSP.

8. Procedure for Possessing and finalizing Audit Objections and Draft Paras.

9. Delegation of Powers.

10. Losses, Frauds and Embezzlements.

11. General Financial Rules

12. Government e-Market (GeM)

13. Classification of Railway Revenue (Earnings)

14. Information Technology in general with specific reference to Railway's IT Applications

15. Taxation matters with special focus on GST & Income Tax

16. Organization of CGA and C&AG

17. Any other topic felt necessary from time to time

Syllabus for Selection from Group 'C' to Group 'B' posts in EDP Centres/IT Department of Railways/Production Units.

Professional Subject:

1. Basics of computing and information Processing

1.1 Business Data Processing

1.1.1. Basic features of off-line systems, transaction processing systems, management information systems, decision support systems. Graphical user interfaces and presentation of information.

1.1.2. Full familiarity with word processor, spreadsheet package, presentation package, email Client and web-browser. Basics of data storage in a database management system.

1.2 Computer organisation and Data structures

1.2.1. Fundamentals of digital logic circuits, binary number system, registers and counters, microprocessor architecture, memory and input/output devices, basics of assembly language.

1.2.2. Data types, stacks, queues, linked lists and other basic data structures, searching and sorting algorithms, and basic idea of complexity.

1.2.3. Basic features of PCs and servers, printers, backup devices.

2. Operating system, DBMS and networking concepts (candidates are expected to specialize in any one area out of Operating Systems, DBMS, or Networking, with a basic idea of all three areas).

2.1. **Basics of operating systems** (proficiency in one Unix flavour or one Windows Server version is expected).

2.1.1. The file system, process management, memory management, syste. administration concepts, networking tools in operating systems, OS level backup and archival, retrieval and disaster management.

2.1.2. Unix flavours, the vi editor, shell programming, basics or grep and awk, user management, security features in Unix, Linux.

2.1.3. The Windows operating system, including basics of Windows server; file management, FAT, FAT32 and NTFS, user management, the active directory, the Windows registry.

2.2. Basics of DBMSs

2.2.1. The relational model, ER model, database normalization concepts.

2.2.2. Basics of SQL, simple queries, joins, nested queries, grouping, query parsing, Indexing concepts.

2.2.3. Database administration, tuning concepts, locking and concurrency control, transaction roll-back and roll forward, database backup and archival.

2.2.4. Proficiency in any one DBMS out of Oracle, Sybase, SQL Server, covering the concepts given above.

2.3 Networking and data communication concepts

2.3.1. Basics of data communication, transmission media, network topologies, the OSI seven layer model, TCP/IP concepts.

2.3.2. LANs and LAN devices, concepts of IEEE 802.3, IEEE802.11b, LAN media, switching, basics of LAN design.

2.3.3. WANs and WAN devices, concepts of routing, switched and packet networks, signal multiplexing methods, network management.

2.3.4. Basics of network security, the Internet, Internet security, viruses, worms and malicious code, intrusion detection.

3. Software development

3.1. **Basics of systems analysis and design**

3.1.1. Modular application design; methods to elicit user requirements; DFDS, ERDS and process mapping methodologies; screen, report and menu design.

3.1.2. Basic idea of component based design, interface design, directory services, web services, object orientation.

3.2. **Application Development** (Proficiency is expected in any one GUI tool, 3GL, or COBOL, with a basic idea of all three)

3.2.1. Using GUI front-end tools, screen building, connecting to back-end databases, and report-writing (any one of VB, D2K/iDS, Power Builder, or other standard GUI tool).

3.2.2. Application development in a structured 3 GL (C, Java, C++, C#) programme constructs in 3 GLs, data types, memory management working with files, working with databases, using standard libraries; the JVM and CLR.

3.2.3. Application development in COBOL, COBOL data structures, file handling in COBOL, sorting and merging, managing databases using COBOL, reporting and screen development.

3.3. **Software engineering concepts**

3.3.1. Software development cycle, basics of software sizing and estimation, software project management, project risk management (basic idea), system audit and security, disaster management strategies.

3.3.1. Basics of software testing and quality assurance, test plans, block box and white box testing, module and integration testing, verification, validation and inspection.

3.3.2. Basics of software testing and quality assurance, test plans, black box and white box testing, module and integration testing, verification, validation and inspection.

3.3.3. Basics of information security, encryption, digital signatures, PKI.

4. **Other Topics in IT (for persons operating in specialized areas)**

4.1. Basics of CAD/CAM

4.1.1. Graphic devices, 2-D geometry and display algorithms, graphic operations, 3-D rendering, animation, graphic standards.

4.1.2. CAD basic, 2-D drafting, surfacing, 3-D solid modelling (proficiency in using a CAD package):

4.1.3. FEA, part programme generation (proficiency in using FEA and part program generation tools.

4.2. Basics of embedded software

4.2.1. Firmware devices, coding and burning, embedded OS, embedded application software, basics of PLCs, ladder logic, using PLC program development tools, data loggers etc.

5. EDP centres in Indian Railways

5.1. Data Centre basics

5.1.1. Data centre basic design features, secure areas and access control, workspace planning; air-conditioning and humidity control; rack-mounted servers and other devices, structured cabling, clean power supply and earthing, UPS basics.

5.1.2. Data centre operations manual, backup strategies, disaster management, basic housekeeping and filing of records.

5.2. IT applications in Indian Railways (as per areas of working of the candidate)

5.2.1. FMIS, Payroll, Workshop system, Inventory control system, Traffic accounts system: detailed knowledge of processes.

5.2.2. SDT applications: PRIME/AFRES, PMIS, MMIS, PAS, FAS, OSS: knowledge of processes.

5.2.3. PRS, UTS, NTES, FOIS: Basic working.

5.2.4. Locally developed applications: for example STORMS, PURSYS, PINS, EASY, PLANS, FACT in RCF; Stores, finance and workshop applications in RWF, DLW, ICF etc.

5.2.5. Other developments on Railways: MIS applications, IR websites, Internet ticketing, IVRS.

6. ICT Infrastructure

6.1 Server Consolidation and Virtualization.

(searchserverervirtualization.techtarget.com)

6.2 Virtualization Systems: VMWare, RHEV, Citrix, Open Stack etc.

(www.redhat.com)

6.3 Software Defined Data Centre and Networks

(https://en.wikipedia.org)

6.4 Cloud Computing and its standards, certification
(www.cio.com)

6.5. Service Oriented Architecture
(http://www.service-architecture.com)

7. **Mobility**

7.1 Mobile Apps Technology Architecture and Programming
(http://www.smartbridge.com)

7.2 Mobile Data Management
(searchmobilecomputing.techtarget.com)

7.3 Mobile Devices Management
(https://www.mobileiron.com)

8. **UIDAI**

8.1 Knowledge of UIDAI Authentication framework
(https://authportal.uidai.gov.in)

8.2 API
(https://www.webopedia.com)

9. **Data Analytics**

9.1 Use of Desktop/Cloud based Data Analytical Tools
(https://www.computerworld.com)

10. **Performance based logistics**

10.1. Parameter definition and determination of SLA's
(https://help.sap.com)

10.2. Measurement using Application Performance Measurement tools
(www.gallop.net)

11. **Knowledge of standards**

11.1. ITIL V3
(www.itilcertification.org)

11.2. ISO 20000
(http://www.iso.org)

11.3. ISO 27000
(www.iso.org)

11.4. ISO 15288
(www.15288.com)

11.5. CMMI

(https://en.wikipedia.org)

12. **Cyber Security Laws and Guidelines**

(http://www.cyberlawsindia.net)

12.1. IT Act

(http://www.itlaw.in)

12.2. MHA Guidelines

(http://mha.nic.in/policy-guidelines)

12.3. NCIIPC Guidelines

(http://cis-india.org)

12.4. CERT-IN Guidelines

(http://www.cert-in.org.in)

12.5. Digital India Policies

(http://meity.gov.in)

12.6. Social Media Guidelines

(http://meity.gov.in)

12.7. Password Policy

(https://mail.nic.in)

12.8. Guidelines for India Government Websites (GIGW)

(http://guidelines.gov.in)

13. **Use of Social Media**

13.1. Twitter

(https://twitter.com/twitter)

13.2. Facebook

(www.facebook.com/)

13.3. You Tube

(https://in.youtube.com/)

13.4. Google+ etc.

(http://www.zigzigger.blogspot.in)

14. **Website Development**

14.1. HTML5

(http://www.w3schools.com)

14.2. XML

(http://www.xmlfiles.com/xml/)

14.3. PHP

(www.php.net)

14.4. Open Source Web CMS Systems: Liferay, Drupal, Joomla, Wordpress.

(https://colorlib.com)